

Developing Gō-Martini Methods for Membrane Protein Conformational Transitions and Lipid Interactions

Conformational transitions are fundamental to protein function, yet characterizing membrane protein dynamics in realistic lipid membranes remains challenging. All-atom molecular dynamics is limited by accessible timescales, while conventional coarse-grained models struggle to describe large-scale conformational changes in explicit membrane environments. To address these challenges, we present a series of methods based on the Gō-Martini model. The Switching Gō-Martini method enables efficient sampling of conformational transition pathways in explicit membranes. Using this method, we revealed the key conformational changes and lipid transport mechanism of the P4-ATPase flippase ATP8A1–CDC50a, and elucidated the substrate translocation of the sphingosine-1-phosphate (S1P) transporter SPNS2. The Multiple-basin Gō-Martini method allows spontaneous transitions between conformational states. Applying this method to the mechanosensitive channels TRAAK and TREK1, we revealed how membrane tension and specific lipids such as POPA and PIP2 regulate the conformational equilibrium. Finally, we developed CTGoMartini, an OpenMM-based Python package, providing a systematic toolkit for studying membrane protein dynamics and protein–lipid interactions.